

Cod (*Gadus morhua*) in divisions 7.e–k (western English Channel and southern Celtic Seas)

ICES advice on fishing opportunities

ICES advises that when the maximum sustainable yield (MSY) approach and precautionary considerations are applied, there should be zero catch in 2027.

ICES notes the existence of a precautionary management plan, developed and adopted by one of the relevant management authorities for this stock.

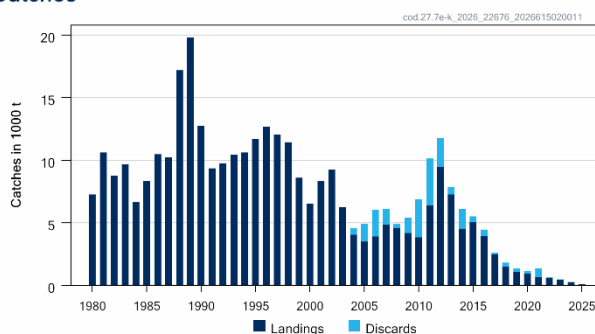
Non-fisheries conservation considerations

Conservation aspects and associated management measures may exist at a national or regional level but were not reviewed by ICES.

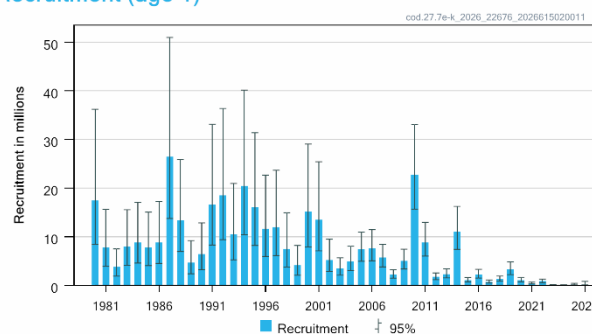
Stock development over time

Fishing pressure on the stock is above F_{MSY} and F_{PA} , and spawning-stock size is below MSY $B_{trigger}$, B_{PA} , and B_{lim} .

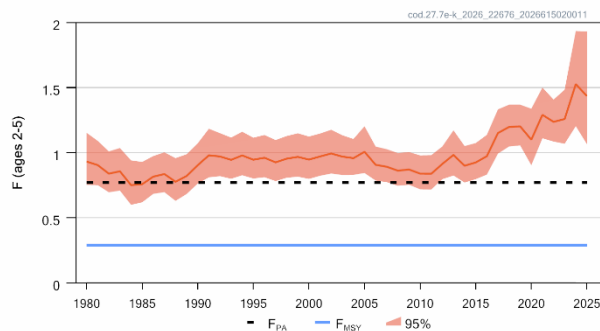
Catches



Recruitment (age 1)



F



SSB

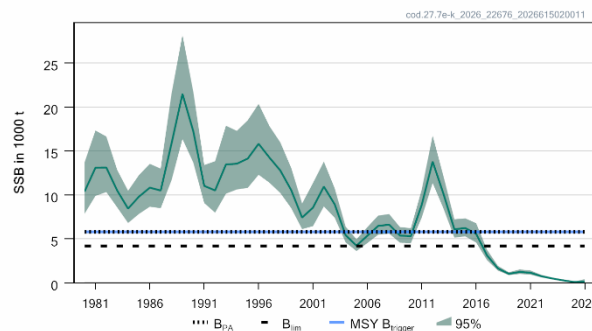


Figure 1

Cod in divisions 7.e–k. Summary of the stock assessment. Discard estimates are available from 2004. The assumed recruitment value for 2026 is shaded in a lighter colour.

Catch scenarios

Table 1 Cod in divisions 7.e–k. Assumptions made for the interim year and in the forecast.

Variable	Value	Notes
$F_{\text{ages 2–5}}$ (2026)	1.41	Average exploitation pattern (2023–2025), scaled to average $F_{\text{ages 2–5}}$ in 2023–2025
Spawning-stock biomass (SSB; 2027)	104	Short-term forecast, in tonnes
$R_{\text{age 1}}$ (2026, 2027)	183	Resampled recruitments from the years 2021–2025 (median shown); in thousands
Total catch (2026)	160	Short-term forecast; in tonnes

Table 2 Cod in divisions 7.e–k. Annual catch scenarios. Weights are in tonnes.

Basis	Total catch (2027)	F_{total} (2027)	Spawning-stock biomass (SSB ; 2028)	% SSB change*	% total allowable catch (TAC) change**	Probability of SSB (2028) < B_{lim} (%) ^{^^}
ICES advice basis						
Maximum sustainable yield (MSY) approach: $F = 0$	0	0	428	310	-100	100
Other scenarios						
$F_{\text{MSY}} \times \text{SSB}_{2027} / \text{MSY } B_{\text{trigger}}$	1	0.0052	426	310	-99	100
EU MAP***: $F_{\text{MSY lower}} \times \text{SSB}_{2027} / \text{MSY } B_{\text{trigger}}$	1	0.003	427	310	-99	100
EU MAP***: $F_{\text{MSY upper}} \times \text{SSB}_{2027} / \text{MSY } B_{\text{trigger}}$	2	0.0074	425	310	-99	100
$F = F_{\text{MSY}}$	67	0.29	333	220	-65	100
$F = F_{\text{MSY lower}}$	41	0.17	369	250	-79	100
$F = F_{\text{MSY upper}}$	90	0.41	300	188	-53	100
$F = F_{\text{PA}}$	148	0.77	222	113	-22	100
$\text{SSB}_{2028} = \text{SSB}_{2027}$	244	1.72	104	0	28	100
$F = F_{2026}$	219	1.41	133	28	15	100
$\text{SSB}_{2028} = B_{\text{lim}}^{\wedge}$						
$\text{SSB}_{2028} = B_{\text{PA}} = \text{MSY } B_{\text{trigger}}^{\wedge}$						

* SSB_{2028} relative to SSB_{2027}

** Total catch in 2027 relative to 2026 TAC (191 tonnes).

*** EU multiannual plan (MAP) for the Western Waters (EU, 2019).

[^] The B_{lim} , B_{PA} , and $\text{MSY } B_{\text{trigger}}$ options were left blank because none of them can be achieved in 2028, even with zero catch in 2027.

^{^^} The probability of SSB being below B_{lim} in 2028. This probability relates to the short-term probability of $\text{SSB} < B_{\text{lim}}$ and is not comparable to the long-term probability of $\text{SSB} < B_{\text{lim}}$; fishing mortalities less than F_{PA} have been shown to have less than 5% probability of the stock being below B_{lim} in the long term when tested through simulations when estimating fishing mortality reference points.

The advice for 2027 is the same as for 2026: zero catch. The zero-catch advice for this stock in 2027 is because the spawning-stock biomass (SSB) is forecast to remain below B_{lim} in 2028 under all catch scenarios.

Basis of the advice

Table 3 Cod in divisions 7.e–k. The basis of the advice.

Advice basis	Maximum sustainable yield (MSY) approach
Management plan	ICES is aware of the multiannual management plan (MAP) that has been adopted by the EU for this stock (EU, 2019) and that ICES considers to be precautionary. There is no agreed shared management plan between the EU and UK for this stock, and ICES provides advice according to ICES MSY approach and precautionary considerations. Catch scenarios consistent with the MAP F_{MSY} ranges are provided.

Quality of the assessment

There is a tendency for the assessment to overestimate SSB and recent recruitment and underestimate F . The final year F has high uncertainty. Small revisions to the last five years input data contributes to higher estimates of recent F . However, this does not impact the advice as the stock is forecast to remain well below B_{lim} .

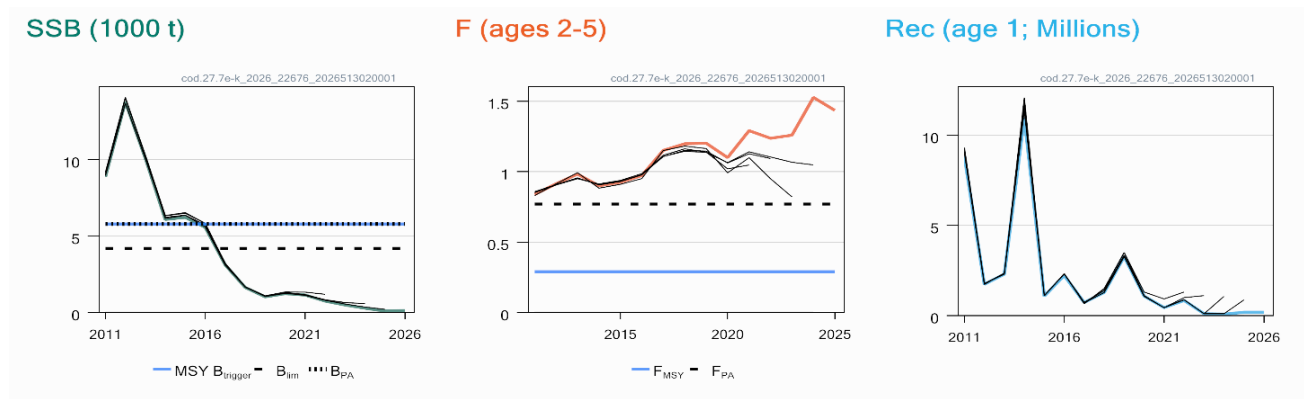


Figure 2 Cod in divisions 7.e-k. Historical assessment results (final-year spawning-stock biomass [SSB] estimate and recruitment assumption included).

Issues relevant for the advice

The assumed recruitments for 2026 and 2027 will contribute 82% of the SSB in 2028 such that the forecasted stock size is strongly dependent on the size of incoming year classes and is therefore subject to increased uncertainty. However, the zero-catch advice is not affected by this uncertainty.

Recent estimates of discard proportions are highly variable because of changing fishery behaviour and limited discard sampling; therefore, no split in forecast discards and landings is provided.

The assessment and the advice are for divisions 7.e-k, including the rectangles 33E2 and 33E3, as cod in these rectangles are considered to be part of the 27.7.e-k cod stock (southern Celtic Seas and western English Channel). Landings data used in the cod 27.7.a assessment explicitly exclude those reported from rectangles 33E2 and 33E3 (Table 8). Data from these two rectangles are incorporated into the advice for cod 27.7.e-k. In 2025, cod landings in 33E2 and 33E3 represented 2.3% of the landings from Division 7.a and constituted 0.7% of the landings from the cod 27.7.e-k stock. The spatial limitation of the advice should be considered when setting total allowable catches (TACs) for the two management areas that span the cod.27.7.a and 27.7.e-k stocks.

Mixed-fisheries considerations

Cod in divisions 7.e-k is caught as part of a mixed fishery, and almost all fisheries operating with demersal gears catch cod (ICES, 2026a).

Reference points

Table 4 Cod in divisions 7.e–k. Reference points, values, and their technical basis.

Framework	Reference point	Value	Technical basis	Source
Maximum sustainable yield (MSY) approach	MSY $B_{trigger}$	5800	B_{PA} ; in tonnes	ICES (2020)
	F_{MSY}	0.29	Segmented regression with B_{lim} (EqSim)	ICES (2020)
Precautionary approach	B_{lim}	4200	B_{loss} , lowest observed spawning-stock biomass (SSB) from which there has been some recovery (2005), rounded value; in tonnes	ICES (2020)
	B_{PA}	5800	$B_{lim} \times 1.4$; in tonnes	ICES (2020)
	F_{PA}	0.77	F_{POS} ; the F that leads to $SSB \geq B_{lim}$ with 95% probability	ICES (2020)
Management plan (MAP)*	MAP B_{lim}	4200	B_{lim} ; in tonnes	EU (2019), ICES (2020)
	MAP F_{MSY}	0.29	F_{MSY}	EU (2019), ICES (2020)
	MAP MSY $B_{trigger}$	5800	MSY $B_{trigger}$; in tonnes	EU (2019), ICES (2020)
	MAP range F_{lower}	0.17	Consistent with ranges resulting in no more than 5% reduction in long-term yield compared with MSY	EU (2019), ICES (2020)
	MAP range F_{upper}	0.41	Consistent with ranges resulting in no more than 5% reduction in long-term yield compared with MSY	EU (2019), ICES (2020)

* EU multiannual plan (MAP) for the Western Waters (EU, 2019).

Basis of the assessment

Table 5 Cod in divisions 7.e–k. Basis of the assessment and advice.

ICES stock data category	1 (ICES, 2025)
Assessment type	Age-based analytical model SAM used in the assessment and the forecast (ICES, 2026b)
Input data	Commercial catches, ages, and length frequencies from sampling by métier; one combined VAST-modelled survey index (combined IGFS-WIBTS-Q4 [G7212] and EVHOE-WIBTS-Q4 [G9527]); one commercial index (FR-OTDEF Q2+3+4 trawlers in divisions 7.e–k); fixed maturity ogive derived from UK-WCGFS survey Q1 data; age-dependent natural mortalities (time invariant; Lorenzen, 1996). Catch weights are from commercial sampling; stock weights use Q1 catch data. Both are averaged over the last three years.
Discards and bycatch	Included in the assessment. Observer-based estimates from 2004; discards and bycatch prior to that (between 1980 and 2003) are model estimated.
Indicators	None
Other information	Benchmarked at WKCELTIC 2020 (ICES, 2020)
Working group	Working Group for the Celtic Seas Ecoregion (WGCSE)

History of the advice, catch, and management

Table 6 Cod in divisions 7.e–k. ICES advice, the agreed total allowable catch (TAC) and ICES estimates of landings and discards. Weights in tonnes.

Year	ICES advice	Landings corresponding to advice	Catch corresponding to advice	Agreed TAC*	ICES landings^^^	ICES discards
1987	Reduce F	< 6400**			10390	
1988	No increase in F ; TAC	7000**			17196	
1989	No increase in F ; TAC	8600**			19804	
1990	No increase in F ; TAC	9200**			12878	
1991	TAC; SSB mean	4500**			9336	
1992	Appropriate to reduce F	-			9747	
1993	20% reduction in F	6500**		17500	10416	
1994	20% reduction in F	5600**		17000	10620	
1995	20% reduction in F	4700***		17000	11710	

Year	ICES advice	Landings corresponding to advice	Catch corresponding to advice	Agreed TAC*	ICES landings ^{^^}	ICES discards
1996	20% reduction in F	4700***		20000	12681	
1997	20% reduction in F	7400***		20000	12028	
1998	10% reduction in F	8800 [^]		20000	11418	
1999	Reduce F below F_{pa}	9200 [^]		19000	8580	
2000	Reduce F below F_{pa}	< 7600 ^{^^}		16000	6539	
2001	40% reduction in F	< 4300 ^{^^}		10500	8316	
2002	45% reduction in F	< 5300 ^{^^}		8700	9239	
2003	60% reduction in F	< 3800 ^{^^}		n/a	6425	
2004	90% reduction in F or management plan	< 700		n/a	4027	543
2005	17% reduction in F	< 5200		6200	3478	1426
2006	No increase in effort (should have been “reduce effort”)	Cannot be estimated		5580	3902	2118
2007	Zero catch	0		4743	4842	1248
2008	Zero catch	0		4316	4577	306
2009	50% reduction in F	< 2600		4023	4187	1229
2010	Substantial catch reduction	-		4023	3831	3040
2011	Catch and effort reduction	-		5379	6376	3749
2012	Maximum sustainable yield (MSY) framework	< 10000		10059	9443	2341
2013	MSY framework	< 10200		10200	7273	562
2014	MSY approach	< 6848		6848	4512	1569
2015	MSY approach	< 4024		5072	5028	483
2016	MSY approach	≤ 3569		4565	3924	525
2017	MSY approach	≤ 1447		2830	2471	134
2018	MSY approach	≤ 3076	≤ 3428	3076	1496	316
2019	MSY approach		0	1610	1051	300
2020	MSY approach		0	805	922	231
2021	MSY approach and precautionary considerations		0	805	629	733
2022	MSY approach and precautionary considerations		0	644	574	86
2023	MSY approach and precautionary considerations		0	644	445	13
2024	MSY approach and precautionary considerations		0	644	249	23
2025	MSY approach and precautionary considerations		0	644	92	11
2026	MSY approach and precautionary considerations		0	191		
2027	MSY approach and precautionary considerations		0			

* TAC covers subareas 7 (except Division 7.a) and 8. From 2009 onwards, the TAC covers divisions 7.b–c and 7.e–k, subareas 8–10, and EU waters of Fishery Committee for the Eastern Central Atlantic (CECAF) Division 34.1.1 (with a separate TAC established for Division 7.d).

** For the division 7.f–g stock component.

*** For the division 7.f–h stock component.

[^] For the division 7.e–h stock component.

^{^^} For the division 7.e–k stock component.

^{^^^} Landings include those taken or reported in rectangles 33E2 and 33E3 since 2004.

History of the catch and landings

Table 7 Cod in divisions 7.e–k. Catch distribution by fleet in 2025, as estimated by ICES.

Catch	Landings				Discards			
103 tonnes	Otter trawls	Seine nets	Beam trawls	Other gears	Otter trawls	Seine nets	Beam trawls	Other gears
	55%	27%	11%	7%	18%	36%	46%	0%
	92 tonnes				11 tonnes			

Table 8 Cod in divisions 7.e–k. History of official landings by country and ICES total estimates of landings and discards. Weight in tonnes.

Year	Belgium	France	Ireland	UK	Others	Total official landings	ICES Landings	ICES Discard estimates	Landings from rectangles 33E2 and 33E3*
1971	807	4382	233	298	138	5858	5782	n/a	
1972	395	3893	275	328	424	5315	4737	n/a	
1973	525	2927	315	196	402	4365	4015	n/a	
1974	197	3023	206	153	324	3903	2898	n/a	
1975	395	3896	259	178	728	5456	3993	n/a	
1976	238	4130	216	139	1195	5918	4818	n/a	
1977	114	3045	183	197	335	3874	3059	n/a	
1978	101	4005	150	182	286	4724	3647	n/a	
1979	119	4437	266	218	0	5040	4650	n/a	
1980	184	6232	708	404	8	7536	7243	n/a	
1981	317	8876	777	681	22	10673	10597	n/a	
1982	214	6637	672	859	18	8400	8766	n/a	
1983	265	5678	887	521	104	7455	9641	n/a	
1984	244	5801	891	688	357	7981	6631	n/a	
1985	476	6330	880	981	217	8884	8317	n/a	
1986	383	9018	890	1038	3	11332	10475	n/a	
1987	226	8609	881	1069	0	10785	10228	n/a	
1988	554	13473	1593	1319	0	16939	17191	n/a	
1989	910	27342	1244	1247	13	30756	19809	n/a	
1990	642	16366	1440	1515	25	19988	12749	n/a	
1991	319	8807	1601	1135	0	11862	9336	n/a	
1992	205	7229	1071	1241	0	9746	9747	n/a	
1993	391	8802	825	964	0	10982	10425	n/a	
1994	397	7925	1472	940	2	10736	10620	n/a	
1995	399	8525	1470	1059	51	11504	11709	n/a	
1996	552	8831	1917	1189	81	12570	12681	n/a	
1997	527	8504	1577	1211	59	11878	12035	n/a	
1998	531	7400	1759	1052	131	10873	11431	n/a	
1999	338	0	1803	961	71	3173	8594	n/a	
2000	212	4439	1529	769	48	6997	6535	n/a	
2001	359	5458	1176	909	52	7954	8309	n/a	
2002	520	4813	824	623	25	6805	9235	n/a	
2003	137	5012	611	355	1	6116	6221	n/a	
2004	153	2310	557	289	1	3310	4027	543	108
2005	187	1706	791	318	1	3003	3478	1426	54
2006	102	2029	936	409	3	3479	3902	2118	103
2007	108	2720	871	435	2	4136	4842	1248	527
2008	66	2201	744	322	0	3332	4577	306	558
2009	49	2104	717	304	4	3178	4187	1229	193
2010	52	1885	974	323	3	3238	3831	3040	143
2011	123	3741	966	473	6	5310	6376	3749	147
2012	290	5287	1593	745	5	7920	9443	2341	85
2013	202	4216	1520	601	1	6540	7273	562	76
2014	141	2166	1284	530	0	4121	4512	1569	24
2015	121	2680	1216	462	4	4482	5028	483	39
2016	97	2166	879	404	1	3546	3924	525	40
2017	82	1457	675	219	0	2433	2471	134	19
2018	49	538	766 ^C	152	0	1505 ^C	1496	316	20
2019	44	412	0 ^C	92	27	575 ^C	1050	300	37
2020	18	425	483	44	52	1022	922	231	71
2021	11	289	300	46	42	687	629	733	52
2022	8	222	312	55	29	626	574	86	30
2023	8	155	209	51	14	438	445	13	19
2024**	6	75	135	28	7	252	249	23	5

Year	Belgium	France	Ireland	UK	Others	Total official landings	ICES Landings	ICES Discard estimates	Landings from rectangles 33E2 and 33E3*
2025**	4	22	55 ^c	11	1 ^c	93 ^c	92	11	<1

* Landings in the southern part of Division 7.a (rectangles 33E2 and 33E3) are included in the assessment and are considered to be part of the stock.

** Preliminary official landings.

^c Incomplete/missing as a result of part of the data being unavailable under data confidentiality clauses.

Summary of the assessment

Table 9 Cod in divisions 7.e–k. Assessment summary. All weights are in tonnes, recruitment in thousands. “Low” and “high” refer to 95% confidence intervals.

Year	Recruitment age 1			Spawning stock biomass (SSB)			Landings	Discards	F age 2–5		
	Low	Value	High	Low	Value	High			Low	Value	High
1980	8443	17486	36213	7862	10381	13706	7243		0.75	0.93	1.15
1981	3931	7836	15617	9890	13091	17326	10597		0.75	0.91	1.09
1982	1911	3784	7490	10326	13111	16648	8766		0.70	0.84	1.01
1983	4120	8006	15560	8586	10503	12849	9641		0.71	0.86	1.04
1984	4582	8843	17068	6808	8444	10475	6631		0.60	0.75	0.94
1985	4051	7812	15066	7839	9802	12258	8317		0.62	0.76	0.93
1986	4516	8832	17274	8661	10828	13536	10475		0.68	0.82	0.97
1987	13745	26474	50992	8489	10501	12991	10228		0.70	0.84	1.00
1988	6951	13415	25891	11692	15879	21567	17191		0.63	0.78	0.96
1989	2392	4690	9194	16367	21482	28196	19809		0.68	0.82	0.99
1990	3210	6419	12833	13664	17220	21701	12749		0.76	0.90	1.07
1991	8304	16582	33113	9078	11029	13398	9336		0.81	0.98	1.18
1992	9380	18471	36371	7985	10512	13838	9747		0.82	0.97	1.15
1993	5228	10467	20958	10142	13464	17874	10425		0.80	0.95	1.12
1994	10407	20442	40154	10623	13551	17286	10620		0.83	0.98	1.16
1995	8188	16034	31400	10801	14132	18490	11709		0.80	0.95	1.12
1996	5923	11590	22679	12295	15818	20352	12681		0.81	0.96	1.14
1997	6089	12009	23685	11387	14243	17817	12035		0.78	0.93	1.10
1998	3778	7505	14908	10217	12773	15968	11431		0.81	0.95	1.13
1999	2085	4134	8194	8494	10537	13071	8594		0.82	0.97	1.15
2000	7912	15163	29060	6110	7425	9022	6535		0.80	0.95	1.12
2001	7100	13436	25426	6451	8561	11362	8309		0.82	0.97	1.15
2002	2842	5200	9514	8689	10948	13794	9235		0.84	0.99	1.18
2003	2097	3449	5674	7405	8918	10739	6221		0.83	0.97	1.13
2004	3025	4926	8022	4602	5424	6394	4027	543	0.83	0.96	1.10
2005	4949	7356	10933	3618	4234	4956	3478	1426	0.84	1.01	1.20
2006	5041	7600	11457	4525	5357	6341	3902	2118	0.79	0.91	1.05
2007	3809	5674	8453	5484	6479	7654	4842	1248	0.77	0.89	1.03
2008	1465	2182	3250	5604	6621	7823	4577	306	0.75	0.86	1.00
2009	3333	4977	7433	4560	5378	6341	4187	1229	0.75	0.87	1.00
2010	15653	22750	33065	4525	5306	6222	3831	3040	0.72	0.84	0.98
2011	6021	8838	12973	7431	8883	10618	6376	3749	0.72	0.84	0.98
2012	1170	1734	2572	11363	13799	16757	9443	2341	0.80	0.91	1.05
2013	1546	2298	3414	8442	10108	12103	7273	562	0.83	0.98	1.17
2014	7421	10980	16246	5139	6094	7228	4512	1569	0.77	0.90	1.05
2015	721	1076	1606	5290	6230	7338	5028	483	0.80	0.93	1.07
2016	1490	2219	3304	4587	5578	6784	3924	525	0.83	0.97	1.14
2017	480	725	1094	2583	3120	3768	2471	134	0.99	1.15	1.33
2018	859	1284	1920	1398	1645	1936	1496	316	1.05	1.20	1.37
2019	2207	3269	4841	879	1031	1210	1050	300	1.06	1.20	1.37
2020	723	1079	1609	1012	1241	1522	922	231	0.90	1.10	1.34
2021	292	442	668	938	1148	1406	629	733	1.11	1.29	1.50
2022	548	835	1273	627	752	903	574	86	1.09	1.24	1.41

Year	Recruitment age 1			Spawning stock biomass (SSB)			Landings	Discards	F age 2–5		
	Low	Value	High	Low	Value	High			Low	Value	High
2023	66	104	163	416	498	596	445	13	1.07	1.26	1.48
2024	47	77	126	229	284	351	249	23	1.20	1.53	1.94
2025	82	183	410	75	99	130	92	11	1.07	1.44	1.93
2026	77	183*	835	33	133	424					

* Resampled recruitments from the years 2021–2025 (median shown)

Sources and references

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